

FROM DATA TO ACTION: DESIGNING HUMAN-CENTERED, IMPACT-DRIVEN DECISION SYSTEMS

Assignment 1

Due Date	Submission
September 4, 2026 11:59PM	Gradescope

This assignment is worth **100 points**, accounting for **15%** of the total course grade. It consists of **3 problems** aligned with techniques/tools covered in **lectures in Weeks 1-2**. Each problem may have one or two components: **written** and **programming**. You are expected to first work through the written questions to build context and conceptual understanding, and then, if applicable, complete the programming part.

Please strictly adhere to the Academic Integrity policies.

In general, the following is required:

- Follow [Submission Instruction](#) (2 points)
- Problem #1: [Café Staffing via MIP](#) (40 points)
- Problem #2: [TikTok Trends via SIR](#) (18 points)
- Problem #3: [Duolingo Template Selection via Multi-Armed Bandits](#) (40 points)

Submission Instruction

Gradescope will have **two submission entries**: one for the programming and the other for the written. Failing to follow the submission instructions will result in a 2-point deduction as well as a delay in the grading process.

Programming Submission

One programming submission attempt should include **all Jupyter Notebooks and, if required, CSV files**. Submission attempts are unlimited. The Autograder relies on file names to run and assign grades. Therefore, please do **NOT rename** the provided start notebooks. If the handout specifies a required name for an output file, you must **use that exact same name** so the Autograder can successfully access it.

Written Submission

The written submission should be **1 single PDF file** containing your answers to all written questions. It must strictly follow this handout template without altering its layout or page structure. In other words, your PDF submission should have the same number of pages and the same alignment as this provided PDF template. Misalignment will cause Gradescope to incorrectly detect answer positions, which you can notice right after submission; as a result, your submission may not be graded correctly by the AI-assisted Autograder.

To ensure page consistency and alignment, you may complete the written questions in one of the following ways:

- type in $\text{\LaTeX}$ using the provided `.tex` file (we recommend using Overleaf)
- write on an electronic handout directly
- print the handout, write by hand, and scan it at Letter size

Each written question is either a multiple-choice question (MCQ) or a short-answer question (SAQ). You may receive partial credit on “Select all that apply” MCQs. Specific answer instructions are as follows:

For “Select one” questions, please fill in the appropriate circle completely:

1. (1 pt) **Select one:** Who teaches this course?

☒ Peter Zhang

☐ Michael Johnson

☐ Jennifer Smith

- If you work in L^AT_EX, simply replace `\choice` with `\CorrectChoice` as your answers. Our pre-defined environment will render PDF correctly.
- If you write on an electronic copy, use **black** color to fill the circle. An overfilled circle is acceptable.
- If you write on a print copy and need to change your answer, you may cross out the previous answer and circle in the new answer:

(1 pt) **Select one:** Who teaches this course?

☒ Peter Zhang

☐ Michael Johnson

☒ Jennifer Smith

For “Select all that apply” questions, please fill in all appropriate squares completely:

2. (2 pts) **Select all that apply:** Which ones are the correct course number?

☒ 94-867

☒ 19-867

☒ 12-768

☐ 12-867

☐ None of the above

- If you work in L^AT_EX, please do **NOT modify** the following line, which ensures square option symbol:
`\checkboxchar{$\Box$} \checkedchar{$\blacksquare$}`
- To change answers on a print copy, you may cross out the previous answer(s) and fill in the new answer(s):

(2 pts) **Select all that apply:** Which ones are the correct course numbers?

☒ 94-867

☒ 19-867

☒ 12-768

☐ 12-867

☒ None of the above

For “Short answer” and “Show your plot” questions, please complete inside the corresponding answer boxes. Two sample answers are shown below: one is acceptable, while the other is NOT.

3. (5 pts) **Short answer:** Use the `lipsum` package to generate a paragraph of Lorem Ipsum text.

Quisque ullamcorper placerat ipsum. Cras nibh. Morbi vel justo vitae lacus tincidunt ultrices. Lorem ipsum dolor sit amet, consectetur adipiscing elit. In hac habitasse platea dictumst. Integer tempus convallis augue. Etiam facilisis. Nunc elementum fermentum wisi. Aenean placerat. Ut imperdiet, enim sed gravida sollicitudin, felis odio placerat quam, ac pulvinar elit purus eget enim. Nunc vitae tortor. Proin tempus nibh sit amet nisl. Vivamus quis tortor vitae risus porta vehicula.

↑ This answer is **acceptable**.

- (5 pts) **Short answer:** Use the `lipsum` package to generate the first paragraph of Lorem Ipsum text.

Nulla malesuada porttitor diam. Donec felis erat, congue non, volutpat at, tincidunt tristique, libero. Vivamus viverra fermentum felis. Donec nonummy pellentesque ante. Phasellus adipiscing semper elit. Proin fermentum massa ac quam. Sed diam turpis, molestie vitae, placerat a, molestie nec, leo. Maecenas lacinia. Nam ipsum ligula, eleifend at, accumsan nec, suscipit a, ipsum. Morbi blandit ligula feugiat magna. Nunc eleifend consequat lorem. Sed lacinia nulla vitae enim. Pellentesque tincidunt purus vel magna. Integer non enim. Praesent euismod nunc eu purus. Donec bibendum quam in tellus. Nullam cursus pulvinar lectus. Donec et mi. Nam vulputate metus eu enim. Vestibulum pellentesque felis eu massa.

↑ This answer is **NOT acceptable**.

For “Show your code” questions, the general requirements stay the same as “Short answer” and “Show your plot”. When asked to show code for a function, your response must include the complete function definition with all lines of code (an example below for your reference). Note that pseudocode will NOT receive any credits.

4. (5 pts) **Show your code:** Complete the `show_all_entry()` function.

```
def show_all_entry(array):
    for r in range(array.shape[0]):
        for c in range(array.shape[1]):
            print(array[r, c])
```

You can use one of the following ways to show code:

- Type code in L^AT_EX. The given answer box is already pre-formatted for coding.
- Insert a screenshot of your code inside the answer box
- Write by hand with clear indents

1 Café Staffing via MIP

Workforce staffing is a critical challenge in the services industry. It involves designing work schedules and assigning personnel shifts to meet fluctuating demand for resources over time. Applications are widespread, including but not limited to telephone operators, hospital nurses, police officers, bus drivers, and restaurant staff. Mathematical optimization provides a systematic approach to solving such business problems.

XCoffee is a rapidly growing café in Pittsburgh. The shop currently employs **six full-time staff members**, each working 5 days per week and 8 hours per day, totaling 40 hours per week. Full-time employees provide their availability for six days (Table 1): 1 for available, 0 for unavailable. The manager arranges the weekly schedule accordingly.

	Mon	Tue	Wed	Thu	Fri	Sat	Sun
Amy	1	1	1	1	1	0	1
Bob	1	1	1	1	1	0	1
Cathy	1	1	1	1	1	1	0
Dan	1	1	1	1	1	1	0
Ed	0	1	1	1	1	1	1
Frank	1	1	1	0	1	1	1

Table 1: Full-time Employee Weekly Availability Data

With recent growth in customer traffic, XCoffee’s manager has identified that existing full-time staff cannot meet peak demand during certain hours. To quantify staffing needs, the café has defined **three standard 4-hour shifts** and developed a demand table (Table 2) specifying the minimum number of workers required for each shift on each day of the week.

	Early shift (7 AM – 11 AM)	Midday shift (11 AM – 3 PM)	Late shift (3 PM – 7 PM)
Mon	5	3	2
Tue	5	3	2
Wed	5	3	2
Thu	5	3	2
Fri	6	4	3
Sat	7	5	4
Sun	6	4	3

Table 2: Weekly Staffing Demand

Each full-time employee can be scheduled for two consecutive shifts, thereby fulfilling the daily 8-hour requirement. Since full-time coverage alone is insufficient to meet the staffing demand outlined in Table 2, the café is considering hiring part-time employees. Each part-time employee will be scheduled for exactly one 4-hour shift per working day. For management efficiency, at least one full-time employee is assigned to every shift to provide oversight and supervision for part-time workers.

The manager needs to **decide which day-shift assignments are given to full-time employees and how many part-time employees are required in each shift.**

The objective is to **minimize the total number of part-time hires.**

(Note: Data for Table 2 and 1 is provided in the handout folder named Data)

1.1 Written

Let's define the mathematical notation and solve this problem as a Mixed-Integer Programming (MIP):

Indices and Sets

- $e \in \mathcal{E} = \{\text{Amy, Bob, Cathy, Dan, Emma, Frank}\}$: full-time employees
- $d \in \mathcal{D} = \{\text{Mon, Tue, Wed, Thu, Fri, Sat, Sun}\}$: working days of a week
- $s \in \mathcal{S} = \{\text{Early, Mid, Late}\}$: 4-hour shifts

Parameters

- **Staffing demand**: required minimum number of workers on day d during shift slot s (refer to Table 2)

$$\text{demand}_{d,s} \in \mathbb{Z}_{\geq 0}$$

- **Full-time employee availability**: 1 if full-time employee e is available on day d ; 0 otherwise (Table 1)

$$\text{availability}_{e,d} \in \{0, 1\}$$

Decision Variables The manager needs to decide, for each 4-hour shift every day, which full-time employee(s) are assigned to work and how many part-time employees should be hired.

- **Full-time staffing**: 1 if full-time employee e is assigned to shift s on day d ; 0 otherwise

$$x_{e,d,s} \in \{0, 1\}$$

- **Part-time hiring**: integer number of part-time workers hired for shift s on day d

$$y_{d,s} \in \mathbb{Z}_{\geq 0}$$

- (a) (2 points) **Short answer:** How many decision variables in total? Show your process.

Objective Function Given the fixed full-time workforce, the objective is to minimize the total number of part-time workers hired during the week.

- (b) (3 points) **Short answer:** Write the **objective function**:

Constraints

- (c) (3 points) **Short answer:** Write the **demand constraint**:

Hint: please use $\geq$ instead $=$; otherwise the model may be unsolvable in Gurobi

- (d) (3 points) **Short answer:** Write the **full-time availability constraint**:

Hint: For each employee, check each shift with his/her daily availability

- (e) (3 points) **Short answer:** Write the **full-time weekly 40-hours constraint**:

With 4-hour shift window, each full-time employee should be scheduled for exactly 10 shifts per week.

- (f) (3 points) **Short answer:** Write the **full-time daily 8-hours constraint**:

With 4-hour shift window, each full-time employee should be scheduled for no more than 2 shifts per day.

- (g) (3 points) **Short answer:** Write the **full-time supervision constraint**:

For each shift, there must be at least 1 full-time employee scheduled to supervise part-time workers.

(h) **Full-time consecutive-shift constraint:**

If full-time employee e is scheduled on day d , then e must be assigned to **2 consecutive shifts** on that day. In other words, e cannot be assigned only to the early (7 AM - 11 AM) and late (3 PM - 7 PM) shifts without covering the midday period. Thus, a valid assignment for e is either **Early + Midday** or **Midday + Late**. Since this constraint is tricky, we have provided the formulas below for your programming.

$$x_{e,d,s=\text{Mid}} \geq x_{e,d,s=\text{Early}} \quad \forall e, \forall d \quad (\text{middle dominates})$$

$$x_{e,d,s=\text{Mid}} \geq x_{e,d,s=\text{Late}} \quad \forall e, \forall d \quad (\text{middle dominates})$$

$$x_{e,d,s=\text{Mid}} \leq x_{e,d,s=\text{Early}} + x_{e,d,s=\text{Late}} \quad \forall e, \forall d \quad (\text{if middle on, at least one neighbor})$$

1.2 Programming

Complete the provided start Jupyter notebook, `D2A_F25_A1_MIP.ipynb`, to solve this optimization problem.

Export the following spreadsheets of your model results as guided in the notebook.

(a) *(10 points)* `full_time_staffing.csv`

(b) *(10 points)* `part_time_hiring.csv`

2 TikTok Trends via SIR

Just as the flu spreads in a population, TikTok trends spread among users through exposure, reposts, and eventual fatigue. Viral growth on TikTok is not random, which follows clear dynamics of discovery, imitation, and decline. We can use the SIR epidemic model to study how a TikTok trend spreads across the user base and to inform platform and creator strategies:

$$S(t+1) - S(t) = -\beta \cdot \frac{S(t)}{N} \cdot I(t) \quad (1)$$

$$I(t+1) - I(t) = \beta \cdot \frac{S(t)}{N} \cdot I(t) - \gamma \cdot I(t) \quad (2)$$

$$R(t+1) - R(t) = \gamma \cdot I(t) \quad (3)$$

- $S(t)$ = users who have not yet seen the trend (susceptible)
- $I(t)$ = users who are actively posting or resharing the trend (infected)
- $R(t)$ = users who are tired of the trend and have moved on (recovered)

2.1 Written

- (a) (3 points) **Select one:** Which of the following best describes “total population” N ?
- ☐ The number of TikTok employees
 - ☐ The total number of viral posts
 - ☐ The number of users who liked a trend
 - ☐ The total number of TikTok users
- (b) (3 points) **Select one:** Which of the following best describes the role of β in the TikTok SIR model?
- ☐ The probability a user stops reposting a trend
 - ☐ The average number of reshares per active user per time step
 - ☐ The number of total TikTok users
 - ☐ The speed of app downloads
- (c) (3 points) **Select one:** Which of the following best describes the role of γ in the TikTok SIR model?
- ☐ The probability that a user first encounters the trend
 - ☐ The average number of reshares per active user
 - ☐ The probability that an active user stops resharing at each time step
 - ☐ The total number of TikTok users in the population
- (d) (3 points) **Select one:** If β is high and γ is low, what happens to the trend?
- ☐ It fizzles quickly
 - ☐ It spreads rapidly and sustains virality
 - ☐ It never catches on
 - ☐ It disappears after one step
- (e) (3 points) **Select one:** If TikTok reduces the visibility of a harmful trend by slowing resharing (e.g., requiring confirmation before reposting), which parameter of the SIR model is most directly affected?
- ☐ The total user population N
 - ☐ The number of recovered users $R(t)$
 - ☐ The transmission rate β
 - ☐ The recovery/fatigue rate γ
- (f) (3 points) **Select one:** If $I(t)$ is growing rapidly from one day to the next, which of the following must be true?
- ☐ γ is very high, meaning users are burning out quickly
 - ☐ New reshares are outpacing fatigue (recoveries)
 - ☐ $R(t)$ is shrinking
 - ☐ The total population N is decreasing

3 Duolingo Template Selection via Multi-Armed Bandits

In this problem, we adapt the business context of the Duolingo case introduced in the lecture. Recall that each Duolingo notification is generated from one template that is selected from a pool. Then the business question is **which template to select** to maximize user engagement.

Duolingo proposed their Sleeping, Recovering Bandit Algorithm in a MAB setup:

$$\pi(a \mid t) = \frac{\exp(s_a^t / \tau)}{\sum_a \exp(s_a^t / \tau)} \quad (4)$$

$$i_t \sim \pi(\cdot \mid t) \quad (5)$$

- Step t : a notification push
- Arm a : a message template
- Policy $\pi(a \mid t)$: probability of selecting action a at timestep t
- Action i_t : an action is sampled from policy distribution $i_t \sim \pi(\cdot \mid t)$
- Arm score s_a^t : adjusted relative reward score with memory decay penalty of arm a at timestep t
- Hyperparameter τ : exploration temperature

We will implement ϵ -Greedy (Equation 6) and Upper Confidence Bound (Equation 7) in this assignment.

$$\epsilon\text{-Greedy:} \quad i_t = \begin{cases} \operatorname{argmax}_a r_a^t & \text{with probability } 1 - \epsilon \\ \text{random action} & \text{with probability } \epsilon \end{cases} \quad (6)$$

$$\text{UCB:} \quad i_t = \operatorname{argmax}_a \left[r_a^t + c \sqrt{\frac{\ln(t)}{n_a^t}} \right] \quad (7)$$

- r_a^t : average reward of arm a received up to timestep t
- n_a^t : number of times arm a selected up to timestep t
- ϵ, c : hyperparameters

3.1 Written

- (a) (5 points) **Select all that apply:** Which of the following statements correctly describe the effect of the softmax exploration temperature parameter τ in policy $\pi(a | t)$ (Equation 4)?
- ☐ Larger τ makes the policy more exploitative, concentrating probability on the top arm.
 - ☐ Smaller τ makes the policy more exploratory, approaching a uniform distribution.
 - ☐ As $\tau \rightarrow \infty$, the policy becomes nearly uniform over all actions.
 - ☐ With $\tau = 1$, probabilities equal the raw scores normalized by their sum.
 - ☐ At each timestep, after calculating $\pi(a | t)$ for each a , select the highest-probability action.
- (b) (3 points) **Select one:** In ϵ -greedy algorithm (Equation 6), what is the main role of the parameter ϵ ?
- ☐ To increase the learning rate of average reward updates
 - ☐ To ensure that each arm is selected exactly the same number of times
 - ☐ To guarantee convergence to the optimal arm after a fixed number of iterations
 - ☐ To balance exploration (trying new template) and exploitation (choosing the best-known one)
- (c) (3 points) **Select one:** Why does UCB (Equation 7) guarantee that each template is selected at least once?
- ☐ Because ϵ forces exploration with probability 0.2
 - ☐ Because templates with $n_a = 0$ have $UCB_a = \infty$
 - ☐ Because r_a starts at 0 for all templates
 - ☐ Because tie-breaking forces fairness

3.2 Programming

With the business context and technical concepts, **you should work through the provided start Jupyter notebook, D2A_F25_A1_MAB.ipynb, before answering the following questions.**

Please note that this programming part does not require any additional spreadsheet submissions; **all responses should be included in this PDF based on your programming implementation.**

- (a) (3 points) **Select one:** With the 100-step initialization, which template should be selected for the next step (i.e., $t = 101$) under a **full exploitation** policy?
- ☐ Template 0
 - ☐ Template 2
 - ☐ Template 5
 - ☐ Template 10

- (b) *(5 points)* **Show your code:** Complete the `update()` function in MAB class.

- (c) *(5 points)* **Show your code:** Complete the `step()` function in `EpsilonGreedy` class.

- (d) *(5 points)* **Show your code:** Complete the `step()` function in UCB class.

- (e) *(6 points)* **Show your plot:** Include your algorithm comparison plot and explain the plot in words. What did you find from the plot? Your plot should:

- have 3 `EpsilonGreedy` lines, 3 `UCB` lines, and 1 `FullRandom` line
- clearly indicate hyperparameters (up to your choice) you have experimented with in the legend

- (f) *(5 points)* **Short answer:** We have explored MAB algorithms as a solution for selecting Duolingo notification templates. How would you revise this problem solution to better reflect real-world business needs? Justify your answer. You may consider aspects such as assumptions, experimental design, evaluation metrics, algorithmic choices, etc.

Deliverables

Congratulations! All of the work is complete. The last step is to submit your deliverables to Gradescope:

- 1 single PDF file to “**Assignment 1 Written**”
- the following files to “**Assignment 1 Programming**”
 - D2A_F25_A1.MIP.ipynb
 - D2A_F25_A1.MAB.ipynb
 - full_time_staffing.csv
 - part_time_hiring.csv